

WASP-12b

R-0570

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_231220 · created 2026-07-24T15:52:31+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460298.9546 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.045
Transit depth (Rp/Rs)^2	2.35 +/- 1.38 [%]
Orbital Inclination (inc)	79.89 +/- 2.94
Airmass coefficient 1 (a1)	0.88 +/- 0.59
Airmass coefficient 2 (a2)	0.11 +/- 0.19
Scatter in the residuals of the lightcurve fit is	70.86 %
Best Comparison Star	None
Optimal Aperture	2.48
Optimal Annulus	7.2
Transit Duration (day)	0.12 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.045 vs 0.1178 (deviation 0.74σ)
Duration fit vs published	0.12 d vs 0.125 d (deviation 0.43σ)
Mid-time O–C	-3.4 min
Depth SNR	3.2
Residual scatter	70.86 %

Light curve

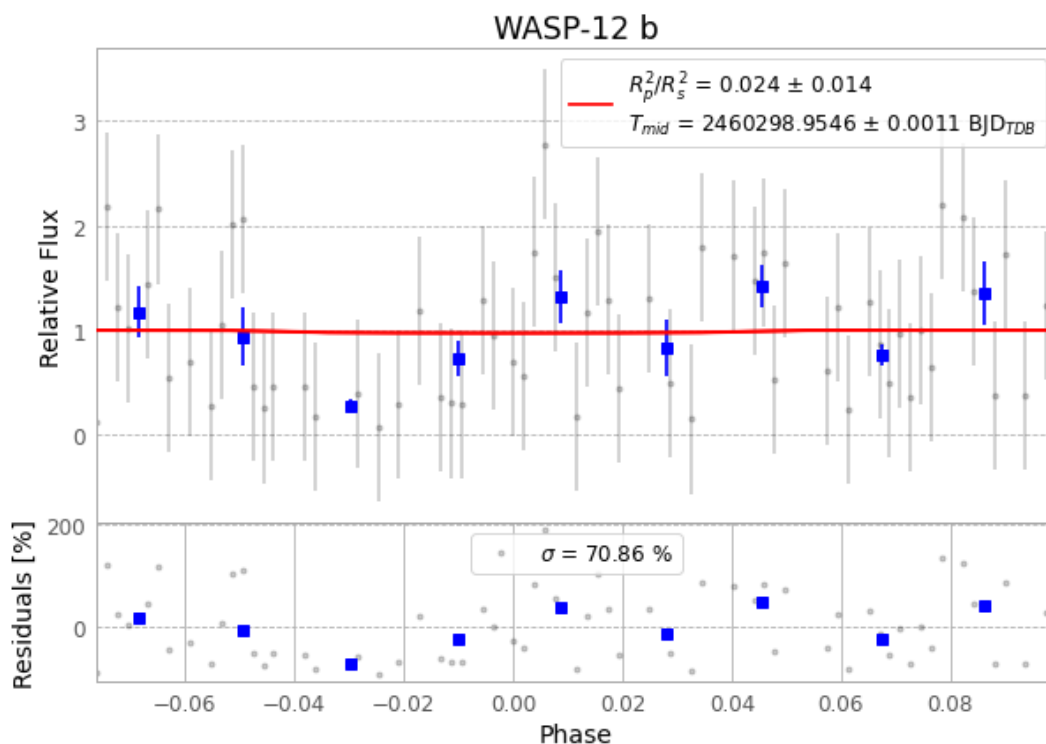


Figure 1 — FinalLightCurve_WASP-12 b_2023-12-20.png (SHA-256 d46b7214bd096146...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [195, 80], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d652d94c44a83c0b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

93 FITS frames (61 MB), WASP-12231220084515.FITS → WASP-12231220132115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-231220.log (4dd8b37e1469...), FinalLightCurve_WASP-12 b_2023-12-20.png (d46b7214bd09...), FinalLightCurve_WASP-12 b_2023-12-20.csv (9ff4a3404527...)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 15:52Z.